

Evaluating Reinforcement Learning Agents in a Custom Game Environment Without Human Input

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Spring 2026

1 Research context and problem statement

Training autonomous agents to perform well in complex environments is one of the main challenges of reinforcement learning. Unlike supervised learning, where a model is trained on labeled examples, reinforcement learning (RL) agents must discover effective strategies through trial and error, guided only by a reward signal. This makes it difficult to evaluate whether an agent has learned a new behavior or is just exploiting patterns specific to its training conditions. The need for researching AI learning behaviors becomes more important as AI is introduced into real world applications, such as self driving cars or medical imaging and diagnostics. These environments prove to be more challenging for evaluating learning behaviors due to their complexity. By training a reinforcement learning agent in a simplified environment (a basic snake game), we will attempt to show evidence that the agent is actually learning over time.

Snake is a well known game that has been used in prior RL research as a controlled testing environment because of its simple reward structure and scalable complexity [3]. Several studies have trained agents on Snake using approaches like deep Q-networks (DQN) [2] and policy gradient methods [4], and open source implementations are available that demonstrate agents reaching high scores. However, much of this prior work focuses on achieving peak performance rather than analyzing how agents learn over time. Our project focuses specifically on how factors such as reward variables and environment conditions affect the learning process of the agent, rather than simply seeking the best outcome.

2 Proposed Solution

To address our research problem, we will design an experimental testbed using a custom implementation of the Snake game environment. The goal is to

train a Proximal Policy Optimization (PPO) agent under varying conditions and analyze its learning behavior using objective metrics.

First, we will develop a custom Snake environment that allows control over factors such as grid size, reward structure, and episode termination conditions. This ensures that the environment is adjustable for experimentation. The sensor data will include relevant information such as the snake’s position, direction, food location, and potential obstacles, enabling the agent to make informed decisions.

Next, we will implement a PPO-based reinforcement learning model using a Gymnasium, which is an API environment built for training agents. The agent will interact with the environment over many training episodes, updating its policy (meaning the agent’s learned decision making strategy) based on observed rewards. PPO is selected due to its stability and effectiveness in discrete action spaces [1] (in this case, forward, left, or right, which are possible moves for the snake), making it well suited for the Snake environment.

A key component of this project is the design of evaluation metrics that do not rely on human judgment. We will track multiple objective measures throughout training, including average score per episode, survival time, reward accumulation, and change in performance. We will also analyze learning curves to determine whether improvements are consistent, and not the result of random fluctuations or overfitting to specific scenarios. Overfitting, when referring to learning agents, is when the agent learns strategies that perform well under the specific conditions it was trained on, but does not reflect genuine, adaptable behavior.

To better understand learning dynamics, we will conduct controlled experiments by varying factors such as reward design and grid size. By adjusting these variables, we can observe how they influence the agent’s ability to learn meaningful strategies. We will compare performance across different configurations to show patterns that indicate genuine learning. For example, a steadily increasing average score combined with reduced variance may suggest stable improvement, while erratic performance may indicate instability or overfitting.

Our project aims to show a method for evaluating the learning outcomes of RL agents based entirely on measurable data. Our findings will show if genuine learning is revealed by training in a simple environment, which could contribute to a broader application in fields where AI learning processes are too complex to observe directly.

3 Evaluation and Implementation Plan

The main question of this project is whether a PPO agent can demonstrate evidence of learning in a custom Snake environment using only objective metrics. Our evaluation plan is designed to answer this question through a series of controlled experiments, each varying one factor at a time while holding all others constant.

We will first establish a baseline by training the PPO agent under a fixed

set of default conditions: a 20x20 grid, a reward structure that awards +10 for eating food, -10 for dying, +1 for moving closer to food, and -1 for moving further away, and a training duration of 1,000,000 timesteps. This baseline will serve as the reference point for all subsequent comparisons. For every training run, we will record the following metrics at regular intervals throughout training: average score per episode (the mean number of food items eaten per episode, averaged over a window of 100 episodes), average survival time (the mean number of timesteps per episode before the snake dies), cumulative reward accumulated per episode, performance variance (the standard deviation of scores across episodes, which shows the stability of the learned policy), and learning curve shape (a visual plot of score over time, used to identify whether improvement is gradual and sustained or erratic).

After establishing the baseline, we will run two more experiments, each changing one variable. First, we will train a second agent using a simpler reward structure (food = +1, death = -1, all other steps = 0) and compare its learning curve against the baseline, testing whether reward shaping changes the learning outcome. Second, we will train a third agent on a smaller 10x10 grid using the baseline reward structure, to examine how environmental complexity affects learning speed and final performance.

We will consider an agent to have genuinely learned if it shows a consistent upward trend in average score over training, achieves a meaningfully higher average score than a random agent, and demonstrates decreasing variance over time, indicating a stable policy. Failure would be shown by a flat or declining learning curve, performance no better than random, or extreme instability throughout training. All training runs will use the same random seed to ensure reproducibility, and results will be visualized as learning curve plots and summarized in a table comparing average final score, average survival time, and performance variance across all three experimental conditions.

3.1 Timeline

The following timeline covers the remaining weeks of the spring semester, from late February through the April 30th final deadline. Buffer time has been built in before the final submission because research and debugging consistently take longer than expected.

- **Weeks 1–2 (Feb 27 – Mar 12): Environment Development.** Complete the custom Snake environment in Python using the Gymnasium API, including game logic, collision detection, food placement, reward calculation, and visual rendering using Pygame. Manually verify the environment works correctly by watching a random agent play.
- **Week 3 (Mar 13 – Mar 19): Training Pipeline.** Set up the training script using Stable Baselines 3 and PPO, configure logging to track metrics throughout training, and run a short test training run (100,000 timesteps) to confirm everything connects correctly.

- **Weeks 4–5 (Mar 20 – Apr 2): Experiments.** Run all three experiments to completion (1,000,000 timesteps each), save all results, and generate learning curve plots.
- **Week 6 (Apr 3 – Apr 9): Analysis.** Analyze results across all three experiments, identify patterns in learning curves, compare final performance metrics, and draw conclusions about which conditions produced the most effective learning.
- **Weeks 7–8 (Apr 10 – Apr 23): Writing and Presentation Preparation.** Write up findings in final report format, design the research presentation summarizing motivation, methods, results, and conclusions, and review and revise both deliverables with research partner.
- **Week 9 (Apr 24 – Apr 30): Final Buffer and Submission.** Reserved for unexpected delays, additional debugging, or rerunning experiments if needed. Final proofreading and submission by April 30th.

References

- [1] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv*, 2017.
- [2] Alessandro Sebastianelli, Massimo Tipaldi, Silvia Liberata Ullo, and Luigi Glielmo. A deep q-learning based approach applied to the snake game. In *29th Mediterranean Conference on Control and Automation (MED)*. IEEE, 2021.
- [3] Md. Rafat Rahman Tushar and Siddique Shahnewaz. A memory efficient deep reinforcement learning approach for snake game autonomous agents, 2023.
- [4] Zhepei Wei, Di Wang, Ming Zhang, Ah-Hwee Tan, Chunyan Miao, and You Zhou. Autonomous agents in snake game via deep reinforcement learning. In *2018 IEEE International Conference on Agents (ICA)*. IEEE, 2018.

3.2 Revisions

We made several revisions since our previous submissions. We edited for brevity and clarity, in accordance with our peer feedback, and added brief definitions for key technical terms. We also added stronger motivation for our research problem, which describes how our project is relevant for real world applications, and added an additional source.